

DESIGN & IMPLEMENTATION REPORT

AlgoNova – AI-Powered DSA & System Design Mastering Platform

1. Project Overview

AlgoNova is an interactive technical-learning and interview-preparation platform combining DSA learning, coding practice, AI-powered interview simulation, System Design practice, resume analysis, and gamified progress tracking.

2. Design Approach

The system follows a decoupled client-server architecture. React and Vite provide the frontend, Node.js and Express.js provide backend services, MongoDB Atlas provides persistent storage, and external services such as Google Gemini and Google OAuth support AI and authentication workflows.

3. Design Changes / Decisions

- Unified learning, coding, interview, System Design and career-readiness platform.
- Interactive DSA visualization using execution states and animation.
- Gemini-based AI evaluation for interviews, System Design, resume analysis, job matching and code feedback.
- Excalidraw-based System Design sandbox for architecture modelling.
- Manual authentication plus Google OAuth with JWT authorization, password hashing, CORS and rate limiting.
- XP, ranks, badges, streaks and leaderboard-based gamification.
- Client-side resume PDF extraction for the ATS workflow.

4. Module-wise Implementation

Module	Implementation Details
AI Interview	Randomized questions, timed assessment, coding response, spoken explanation, proctoring signals and AI evaluation.
System Design	Excalidraw canvas for components/connections; architecture representation is evaluated for scalability, fault tolerance and bottlenecks.
Resume ATS & Job Matcher	Client-side PDF extraction, rule-based ATS analysis, Gemini linguistic critique and skill-based job matching.
DSA Visualizer	Pre-built algorithms and Bring Your Own Code workflow with execution-state mapping and graphical numerical-array visualization.
Practice HQ	Coding problem interface, source-code submission, execution results and AI-assisted complexity/code-quality feedback.
Gamified Progress	Activity results contribute to XP, ranks, badges, streaks and leaderboard information.
Authentication & Security	Manual login/registration and Google OAuth, JWT authorization, bcryptjs, rate limiting and CORS controls.

5. Technology Implementation

The documented stack uses React 19 with Vite and Tailwind CSS, Node.js and Express.js, MongoDB Atlas with Mongoose, Google Gemini, Monaco Editor, Excalidraw, pdfjs-dist, authentication/security libraries, and supporting services for email, animation, and deployment.

6. System Architecture

The architecture is organized into presentation, application, and data/external-service layers. The presentation layer contains the React interface and interactive tools. The application layer contains Express APIs, business logic, authentication/security middleware and feature services. MongoDB Atlas and external services form the data and integration layer.

7. Prototype Screenshots

The assignment requires screenshots of working screens. Actual application screenshots were not present in the supplied source material for this document, so no artificial screenshots are included. Insert screenshots from the deployed/local prototype for: Login/Registration, Dashboard, DSA Visualizer, AI Interview, System Design Sandbox, Resume ATS, and Practice HQ.

8. Conclusion

The prototype establishes a modular full-stack architecture integrating interactive technical learning, AI-assisted assessment, coding practice, System Design, resume analysis and gamification within a common application.